LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: INT396	Unit: 2
Course Title: UNSUPERVISED LEARNING	Unit Title: Partition-Based Clustering
Course Outcome: CO2	Unit Mapped RBT: L4 (Analyze)
Paper: Sample Multiple Choice Questions - ANSWER KEY	Total Questions: 40
Maximum Marks: 40 x 1 = 40	Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. State what the letters WCSS stand for in the reporting of a k-Means result.

A) Within-Cluster Sum of Squares <-- correct
B) Weighted Cluster Similarity Score
C) Whole Cluster Separation Score
D) Within-Cluster Silhouette Sum

Answer: A. WCSS is the sum of squared distances from each record to its own cluster centre. It is the same quantity that is reported as inertia.

Q2. Identify the shape of the region of space that a single k-Means cluster occupies.

A) A ring around one centre
B) A convex region around one centre <-- correct
C) A curved band of any shape
D) A set of disconnected patches

Answer: B. A cluster is the set of points nearer to its own centre than to any other, which is a convex cell. This is why non-convex groups cannot be recovered.

Q3. Name the two initialisation strategies given on the syllabus for choosing the starting centres of k-Means.

A) Random initialisation and the elbow method
B) The elbow method and the silhouette method
C) Random initialisation and k-Means++ <-- correct
D) Standardisation and min-max scaling

Answer: C. The centres may be drawn at random, or spread out by k-Means++, which picks each new centre with probability proportional to its squared distance from the nearest existing one.

Q4. Identify what the abbreviation PAM stands for in the name of the k-Medoids algorithm.

A) Partitioning And Merging
B) Proximity Assignment Method
C) Partition Average Minimisation
D) Partitioning Around Medoids <-- correct

Answer: D. PAM is the classical k-Medoids procedure. Its centre is a medoid, an actual record, rather than a computed mean.

Q5. Identify the quantity that k-Medoids sets out to minimise.

A) The sum of the squared distances to the mean
B) The sum of the distances to the medoid <-- correct
C) The number of records in the largest cluster
D) The distance between the two nearest medoids

Answer: B. k-Medoids minimises unsquared distances to an actual record. Squaring is what makes the k-Means objective so sensitive to an extreme value.

Q6. State the inertia of a clustering in which k equals the number of distinct records.

A) One
B) The number of records
C) It cannot be computed

D) Zero <-- correct

Answer: D. Every record becomes its own centre, so every distance to a centre is zero. This is why inertia cannot simply be minimised to choose k.

Q7. Explain why a k-Means centroid need not coincide with any record of the dataset.

A) It is an average, not a chosen record <-- correct

B) It is always placed at the origin

C) It is chosen before the data is read

D) It is the record furthest from the rest

Answer: A. The centre is the arithmetic mean of the assigned points, and the mean of several records is usually a position where no record actually sits.

Q8. Distinguish the k-Means cost from the k-Medoids cost by the power of distance each one uses.

A) k-Medoids squares the distances, k-Means does not

B) Both of the two square the distances used

C) k-Means squares the distances, k-Medoids does not <-- correct

D) Neither of the two squares the distances used

Answer: C. Squaring magnifies a large gap far more than a small one, which is precisely why the mean-based objective is dragged about by extreme records.

Q9. Interpret a negative silhouette coefficient reported for a record.

A) It sits exactly at the centre of its cluster

B) It is equidistant from every cluster centre

C) It sits nearer to a neighbouring cluster than its own <-- correct

D) It has been excluded from every cluster

Answer: C. The silhouette is negative when the mean distance to the record's own cluster exceeds the mean distance to the nearest other one, which indicates a misplacement.

Q10. Describe what min-max scaling produces from a numeric column.

A) Values lying between 0 and 1 <-- correct

B) Values with mean 0 and variance 1

C) Values lying between -1 and +1

D) Values converted into whole numbers

Answer: A. Min-max scaling subtracts the minimum and divides by the range, so the smallest value becomes 0 and the largest becomes 1.

Q11. Predict what must happen if k is set larger than the number of distinct records in the table.

A) The algorithm splits records into halves

B) The extra centres are placed at the origin

C) Every cluster receives the same record

D) At least one cluster must be left empty <-- correct

Answer: D. There are not enough distinct positions to give every centre a point of its own, so implementations either drop or reinitialise the empty clusters.

Q12. Recognize the situation described: during an iteration one of the centres ends up with no points assigned to it at all.

A) A converged cluster, which is left alone

B) An empty cluster, which must be reinitialised <-- correct

C) A noise cluster, which is reported as noise

D) A merged cluster, which absorbs its neighbour

Answer: B. An empty cluster has no mean to compute. Implementations restart that centre, usually at the point furthest from its current centre.

Q13. Explain why an inertia value from a scaled table cannot be compared with one from the same table unscaled.

A) Inertia is undefined once a table is scaled

B) Inertia always falls to zero after scaling

C) Inertia is computed from the labels, not features

D) Inertia is expressed in the units of the features <-- correct

Answer: D. Rescaling the columns rescales every squared distance, so the two numbers are on different scales and their comparison says nothing about quality.

Q14. Distinguish the within-cluster sum of squares from the between-cluster sum of squares.

A) Within measures separation, between measures compactness

B) Both of the two measure the compactness of clusters

C) Within measures compactness, between measures separation <-- correct

D) Both of the two measure the separation of clusters

Answer: C. The within-cluster term is the inertia and falls as clusters tighten; the between-cluster term grows as the centres move apart. Good clusterings do both.

Q15. Explain why MiniBatch k-Means normally finishes with slightly higher inertia than full k-Means.

A) It uses a different distance measure entirely

B) Its updates are computed from a noisy sample <-- correct

C) It always stops after a single iteration

D) It refuses to move a centre once it is set

Answer: B. Each update sees only a small batch, so the centres are nudged by an estimate rather than the exact mean. The result is close but not quite as tight.

Q16. Interpret a Davies-Bouldin Index that is very close to zero.

A) The clusters are compact and well separated <-- correct

B) The clusters overlap almost completely

C) The clustering has failed to converge

D) The number of clusters is far too large

Answer: A. The index averages the worst ratio of within-cluster scatter to between-cluster distance. A value near zero means scatter is small relative to separation.

Q17. Predict the effect on k-Means of scaling only one of the two numeric columns.

A) Both columns then contribute equally

B) The column left unscaled continues to dominate <-- correct

C) The scaled column takes over the result

D) The algorithm can no longer be run

Answer: B. Scaling is only useful when it puts the columns on a comparable footing. Scaling one of them leaves the mismatch, and often makes it worse.

Q18. Describe what k-Means assumes about the relative sizes and densities of the groups it looks for.

A) That they are of broadly similar size and spread <-- correct

B) That one group is far larger than the rest

C) That every group has exactly the same shape

D) That the groups may overlap without limit

Answer: A. Because assignment is by nearest centre, a large diffuse group tends to be split while a small tight one nearby is absorbed.

Q19. Solve the following: a cluster holds the two-dimensional points (1, 2), (3, 4) and (5, 6). Determine its centroid.

A) (2, 3)

B) (4, 5)

C) (9, 12)

D) (3, 4) <-- correct

Answer: D. The centroid is the mean of each coordinate: $((1+3+5)/3, (2+4+6)/3) = (3, 4)$.

Q20. Compute the inertia of a one-dimensional cluster holding 4, 6 and 8 whose centre lies at 6.

A) 4

B) 6

C) 8 <-- correct

D) 12

Answer: C. Inertia is the sum of squared distances to the centre: $(4-6)^2 + (6-6)^2 + (8-6)^2 = 4 + 0 + 4 = 8$.

Q21. Apply the silhouette definition to a record whose mean distance to its own cluster is 3 and to the nearest other cluster is 5.

A) 0.4 <-- correct

B) 0.6

C) 1.7

D) 0.2

Answer: A. The silhouette is $(b - a) / \max(a, b) = (5 - 3) / 5 = 0.4$, a moderately well-placed record.

Q22. Use the k-Medoids cost to determine the medoid of the one-dimensional cluster holding 2, 3 and 10.

A) 2

B) 10

C) 3 <-- correct

D) 5

Answer: C. Total distance from 2 is $1 + 8 = 9$; from 3 it is $1 + 7 = 8$; from 10 it is $8 + 7 = 15$. The point 3 gives the lowest cost.

Q23. Solve the following: a feature has mean 100 and standard deviation 20. Determine the standardised value of an observation equal to 70.

A) -0.5

B) -1.5 <-- correct

C) 1.5

D) -3.0

Answer: B. The z-score is $(70 - 100) / 20 = -30 / 20 = -1.5$, so the observation lies one and a half standard deviations below the mean.

Q24. Compute the number of point-to-centre distance calculations made by one assignment step on 2,000 records with k equal to 5.

A) 2,005

B) 400

C) 25,000

D) 10,000 <-- correct

Answer: D. Every record is compared with every centre, so the count is $2{,}000 \times 5 = 10{,}000$.

Q25. Use min-max scaling to determine the scaled value of 30 in a column whose minimum is 10 and maximum is 50.

A) 0.3

B) 0.6

C) 0.5 <-- correct

D) 0.2

Answer: C. Min-max scaling gives $(30 - 10) / (50 - 10) = 20 / 40 = 0.5$.

Q26. Apply the elbow method to the following inertia readings and determine the value of k to report: k = 1 gives 1200, k = 2 gives 500, k = 3 gives 200, k = 4 gives 180, k = 5 gives 172.

A) 2

B) 4

C) 5

D) 3 <-- correct

Answer: D. The falls are 700, 300, 20 and 8. The curve bends sharply after k = 3, beyond which extra clusters buy almost nothing.

Q27. Examine the following and select the correct explanation: a k-Means run on a table of 500 records reports an inertia of exactly zero.

A) Every record has been made a cluster of its own <-- correct
B) The features were standardised beforehand
C) The algorithm stopped before converging
D) The distance measure was chosen wrongly
Answer: A. Inertia is a sum of squared distances to the centres, and it can only vanish when every point sits exactly on a centre.

Q28. Solve the following: a two-dimensional cluster holds the points (0, 0) and (4, 6). Determine its centroid.

A) (4, 6)
B) (2, 3) <-- correct
C) (0, 0)
D) (2, 6)
Answer: B. The centroid is the mean of the coordinates: ((0+4)/2, (0+6)/2) = (2, 3).

Q29. Apply standardisation to a numeric column and determine the mean and standard deviation of the result.

A) Mean 1 and standard deviation 0
B) Mean 0 and standard deviation 0
C) Mean 0.5 and standard deviation 0.5
D) Mean 0 and standard deviation 1 <-- correct
Answer: D. Standardisation subtracts the mean and divides by the standard deviation, which by construction centres the column at 0 with unit spread.

Q30. Apply the k-Means++ rule to two candidate points lying at distances 3 and 9 from the centre already chosen, and determine the probability that the point at distance 9 is selected next.

A) 0.75
B) 0.9 <-- correct
C) 0.5
D) 0.6
Answer: B. Selection probability is proportional to the squared distance: 81 / (81 + 9) = 81 / 90 = 0.9.

Q31. Examine the following and select the correct explanation: after the analyst standardises the table, the k-Means clusters change completely.

A) The number of clusters was silently increased
B) Standardisation removes records from the table
C) Every feature now contributes comparably to the distance <-- correct
D) The algorithm switched to a different objective
Answer: C. Before scaling, the widest-ranging column decided the distances almost alone. Afterwards all of them count, so the geometry, and the partition, is different.

Q32. Compare one iteration of full k-Means with one iteration of MiniBatch k-Means on a dataset of one million records.

A) MiniBatch touches only the records of its batch <-- correct
B) MiniBatch touches every record twice over
C) Both of the two touch every record once
D) MiniBatch touches more records than full k-Means
Answer: A. The cost of a MiniBatch iteration is set by the batch size rather than the size of the dataset, which is what makes it usable at this scale.

Q33. Analyze why the k-Means objective is certain to fall at every step yet is not certain to reach its lowest possible value.

A) Each step reduces inertia but only a local minimum is reached <-- correct
B) Each step raises inertia until convergence is forced
C) The objective is convex, so the minimum is always found
D) The objective is fixed once the centres are initialised
Answer: A. Assignment and update can each only lower the inertia, so the run converges; but the surface has many valleys and the starting point decides which one is entered.

Q34. Differentiate the effect of a single extreme outlier on a k-Means centroid from its effect on a k-Medoids medoid.

A) The medoid is pulled towards it, the centroid is not
B) Both of the two are pulled towards it equally
C) Neither of the two is affected by an outlier
D) The centroid is pulled towards it, the medoid is not <-- correct
Answer: D. A mean has no resistance to an extreme value, whereas a medoid must remain one of the actual records and therefore stays with the bulk of the cluster.

Q35. Infer what it indicates when the silhouette score falls steadily as k is raised from 2 upwards.

A) The data holds a very large number of groups
B) The data holds little genuine group structure <-- correct
C) The features have all been standardised twice
D) The algorithm has failed to converge at any k
Answer: B. Splitting further keeps making the assignments more borderline. When no value of k improves the score, the evidence for real clusters is weak.

Q36. Distinguish what inertia tells you about a partition from what the silhouette coefficient tells you.

A) Inertia reports separation only, silhouette reports compactness
B) Both of the two report compactness and separation
C) Inertia reports compactness only, silhouette reports separation too <-- correct

D) Both of the two report compactness and nothing else

Answer: C. Inertia measures distance to a record's own centre and says nothing about the neighbours. The silhouette compares the two, so it can fall when clusters overlap.

Q37. Analyze the consequence of clustering a table in which one informative column has accidentally been included twice.

A) The duplicate column is detected and removed

B) The distances between records become zero

C) That quantity counts twice in every distance <-- correct

D) The number of clusters is automatically doubled

Answer: C. The duplicated feature contributes two identical squared terms, doubling its weight in the distance and quietly biasing the partition towards it.

Q38. Infer why the elbow method and the silhouette score can point to different values of k on the same data.

A) One of the two is always computed incorrectly

B) They measure different properties of a partition <-- correct

C) They can only differ when the data is unscaled

D) The silhouette is undefined for small values of k

Answer: B. The elbow watches how fast compactness improves; the silhouette weighs compactness against separation. Genuine disagreement is common and must be judged, not ignored.

Q39. Relate the number of restarts used by a k-Means run to how much the reported partition can be trusted.

A) More restarts guarantee the global minimum is found

B) More restarts reduce the number of clusters needed

C) More restarts remove the need to scale the features

D) More restarts make an unlucky single start less likely to survive <-- correct

Answer: D. Keeping the best of many differently initialised runs raises the chance that the reported solution is a good local minimum, though it can never prove it is the best one.

Q40. Differentiate a hard assignment from a soft assignment by what each conveys about a record lying on a boundary.

A) The soft assignment shows the record is borderline, the hard one hides it <-- correct

B) The hard assignment shows the record is borderline

C) Both of the two show that the record is borderline

D) Neither of the two carries information about boundaries

Answer: A. A hard label treats a boundary record with the same confidence as one at the centre of a cluster. Membership weights make the uncertainty visible.

----- End of Sample Question Set -----